# Aurea Soriano-Vargas

*PhD in Computer Science*

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| **Current Position** | **Senior Researcher** | *Institute of Computing - State University of Campinas (UNICAMP)* |
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| **Date of Birth** | December $21^{st}$, 1987 | |
| **Citizenship** | Peruvian - **DNI** 44774893 | ***Brazilian RNE*** *- V716555-T* |
| **CIP** | 214862 | |
| **RENACYT** | P0017484 | *Carlos Monge Medrano IV* |
| **SBC** | 43741 | *Sociedade Brasileira de Computação* |
| **ORCID** | 0000-0002-8429-4119 | ***IEEE:*** *94272623* |

## About me

Expert in Visual Analytics, Information Visualization, and Machine Learning with about 10 years of experience in the area. She is currently a senior researcher in Artificial Intelligence and coordinator of Hub Viva Bem at the Recod.ai laboratory, Institute of Computing, University of Campinas and a member of the IEEE Signal Processing Society. She participated in two postdoctoral projects from 2018 to 2021. She earned a doctorate and a master's degree in Computer Science from the University of São Paulo (USP), Brazil in 2018 and 2013, respectively. She has previously worked as a junior researcher at the University of California, Davis (2015-2016). She has experience in projects with innovative solutions that combine Machine Learning and Information Visualization in time series from different domains, such as musical composition, GPS signals, stock market data, crime statistics, air quality, and oil production data and reservoir simulation data.

**Research Interests:** Artificial Intelligence, Machine Learning, Deep Learning, Information Visualization, Visual Analytics, and Image Processing.

## Education

2013–2018 **PhD in Computer Science**, *Institute of Mathematics and Computer Sciences (ICMC), University of Sao Paulo (USP)*, Sao Paulo, Brazil, **FAPESP fellowship**. Advisor: Dr. Maria Cristina Ferreira de Oliveira - ICMC/USP - Brazil. **GPA: 4.0/4.0**.

2011–2013 **MSc in Computer Science**, *Institute of Mathematics and Computer Sciences (ICMC), University of Sao Paulo (USP)*, Sao Paulo, Brazil, **CNPq and FAPESP fellowships**. Advisor: Dr. Maria Cristina Ferreira de Oliveira - ICMC/USP - Brazil. **GPA: 3.71/4.0**.

2015–2016 **Visiting Scholar in Computer Science**, *Institute for Data Analysis and Visualization (IDAV), University of California, Davis*, California, USA, **FAPESP fellowship**. Supervisor: Dr. Bernd Hamann.

2004–2009 **Bachelor in Informatics Engineering**, *Catholic San Pablo University (UCSP)*, Arequipa, Peru, **University fellowship**. **Graduated with honors. First rank in my class**. Advisor: Dr. Eduardo Tejada Gamero. **GPA: 3.91/4.0**..

## Projects

***Current project***

Senior Researcher - Coordinator — *Viva-bem: Artificial Intelligence Hub in Health and Wellness. State University of Campinas (UNICAMP), Campinas, Brazil.*

Postdoc — *ShellML: Machine-Learning for Analysis of Complex Production Data in a Pre-Salt Carbonate Field Project. State University of Campinas (UNICAMP), Campinas, Brazil.*

Postdoc — *Shell4D: Similarity Exploration to Measure the Misfit between 4D Seismic Data and Numerical Reservoir Models. State University of Campinas (UNICAMP), Campinas, Brazil.*

PhD — *Visualization to Support the Identification of Relevant Attributes in Multidimensional Data Time-Varying. University of São Paulo (ICMC), São Carlos, Brazil.*

MSc — *Alternative Visual Metaphors for Layouts Generated by Multidimensional Projections:A Case Study in Visualization of Music. University of São Paulo (ICMC), São Carlos, Brazil.*

Engineering — *MIDI music visualization and analysis of musical features with traditional classifiers. San Pablo Catholic University, Arequipa, Peru.*

*(Project to obtain license to work in Peruvian territory.)*

BSc — *Generalized Wavelet Product Integral for Rendering Dynamic of Shiny Objects. San Pablo Catholic University, Arequipa, Peru.*

## Honors, Awards and Scholarships

2019-2021 Postdoctoral Scholarship *FUNCAMP/SHELL, Brazil*

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| Nov 2019 | Khipu - Latin American Meeting In Artificial Intelligence Scholarship | *Khipu, Uruguay* |
| May 2019 | Summer School on Biometrics and Forensics Scholarship | *IDENTITY/European Union, Italy* |
| 2018-2019 | Postdoctoral Scholarship | *FUNCAMP/SHELL, Brazil* |
| 2015-2016 | Research Internships Abroad (BEPE) - Scholarship | *FAPESP, Brazil* |
| 2013-2017 | PhD - Scholarship | *FAPESP, Brazil* |
| 2011-2013 | MSc - Scholarship | *CNPQ and FAPESP, Brazil* |
| 2010–2010 | Fellowship in Information Technology | *Southern Copper Corporation, Peru* |
| 2004–2008 | BSc - Scholarship | *Catholic San Pablo University, Peru* |
| Mar 2018 | PhD Thesis approved unanimously with 2 journal publications | *University of Sao Paulo, Brazil* |
| Jan 2015 | Thesis approved unanimously, public mention and recommendation to be published | *Catholic San Pablo University* |
| May 2013 | Master's Thesis approved unanimously with 2 conference publications | *University of Sao Paulo, Brazil* |
| 2004-2008 | First Rank of my class | *Catholic San Pablo University* |
| 2006 | Member of the winning group of the ACM programming contest of Catholic San Pablo University | *Catholic San Pablo University* |

## Continuing Education

2022-I **Jira, Confluence and other software engineering tools**, *Samsung Brasil*.

2021-I **Pedagogy and Didactic in Higher Education**, *National University of San Agustin (UNSA)*.

2021-I **Evaluation of scientific activity: objective use of metrics and indicators**, *Clarivate LatAm*.

2021-I **Resilience and Emotion Control oriented to Professors**, *Department of Psychology of the Catholic San Pablo University*.

2020-I **Webinar - Machine Learning Applications in Well Data**, *European Association of Geoscientists and Engineers (EAGE)*, Authenticity 3676893.3960967.4.6.878439938138768.

2019-II **School of Machine Learning in 2nd Latin American Meeting In Artificial Intelligence (Khipu)**, *Montevideo, Uruguay*, Khipu scholarship.

2019-II **Building Resilient Streaming Systems on Google Cloud Platform**, *Coursera*, License WKCMLFVBGXBK.

2019-II **Serverless Machine Learning with Tensorflow on Google Cloud Platform**, *Coursera*, License B3SVJGBK4QUQ.

2019-I **Serverless Data Analysis with Google BigQuery and Cloud Dataflow**, *Coursera*, License LUJ6TGNGMWY7.

2019-I **Leveraging Unstructured Data with Cloud Dataproc on Google Cloud Platform**, *Coursera*, License W69P2ZFVNFWX.

2019-I **Google Cloud Platform Big Data and Machine Learning Fundamentals**, *Coursera*, License EMHREA8X5RJJ.

2019-I **16th Int. Summer School on Biometrics and Forensics**, *Alghero, Italy*, IDENTITY scholarship.

2018-II **Machine Learning by Andrew Ng**, *Stanford University/Coursera*, License 52YMCQ4GVJD3.

2018-I **2nd School of Researchers**, *Campus USP Sao Carlos*, Databases web of sciente (wos),Orcid (open researcher and contributor id), Research Project, High Impact Science, Search Data Management, Scientific Communication in the Preprints Era, Start (State of the Art through Systematic Review), Theoretical Review of Scientific Methodology, Endnote (Bibliographic Reference Manager), Good Practice in Research, Peer Review Process, Scientific Writing and Publishing of scientific articles.

2018-I **IX Regional School of Informatics - SP / WEST**, *Brazilian Computer Society*, Deep Learning: Introduction and Applications. Techniques of attacks and countermeasures in Information Security.

2017-I **Advanced Big Data School**, *University of Sao Paulo*, Sao Carlos-Brazil, Complex Data Management in Large Scale (SGBDR, NoSQL, Map Reduce, Hadoop, Business Intelligence and Big Data, Big Data warehousing, Hadoop and Apache Hive), Deep Learning (Classification, Location and Segmentation using Deep Nets, Sequence Modeling with Recurrent Neural Networks (RNNs), RNN architectures, Deep Neural Networks for Vision, Tensor Decomposition for Visual Question Answering), Data Mining in Complex Networks (Data Mining and Spatial Analysis, Recommender algorithms, Prequential evaluation and online evaluation, Multidimensional recommendation, Forgetting - Incremental Models), Data visualization (Point-centered Visualization, Attribute-centered Visualization, Relationship-centered Visualization, Visualization and Mining, Similarity, Graph Analytics, Multiple-Perspective Visual Analytics), Multimedia Data Processing (Information Retrieval, Multimedia Data Processing), Machine Learning (Statistical Learning Theory) .

2017-I **Academic Oral Presentations in English**, *University of Sao Paulo*, Sao Carlos-Brazil.

2015-II **Scientific Writing in English**, *University of Sao Paulo*, Sao Carlos-Brazil.

2015-I **Intensive English course**, *Excellent Global*, Sao Carlos-Brazil.

2013-II **Course of Inference via bootstrap and applications**, *University of Sao Paulo*, Sao Carlos-Brazil.

2010 **Course of Advanced Technical English**, *Langrow*, Cuajone-Peru, Course offered to employees of the Information Technology area of Southern Peru Copper Corporation.

2008 **Course of English as a second language - Advanced level**, *Language Center of the Catholic San Pablo University*, Arequipa-Peru.

2007 **Course of Java Programming - Basic and Intermediate Levels**, *Del Sur Institute*, Arequipa-Peru, High grade.

## Experience

2022-I – Currently **Senior Researcher - Coordinator**, *State University of Campinas (UNICAMP)*.
Data Mining, Machine Learning, Information Visualization in Artificial Intelligence Hub in Health and Wellness.

2022-II **Virtual Professor**, *Data Science Research Perú*, Fundamentals of Data Analysis.

2022-I **Computer science major reviewer**, *Universidad Peruana de Ciencias Aplicadas (UPC)*.

2022-I **Virtual Professor**, *Data Science Research Perú*, Fundamentals of Data Analysis.

2022-I **Invited Professor**, *National University of San Agustin*, Design and Analysis of Algorithms.

2019-II – 2021-II **Postdoctoral research fellow**, *State University of Campinas (UNICAMP)*, Campinas-Brazil. Supervisor: Dr. Anderson Rocha.
Data Mining, Machine Learning, Information Visualization in Machine-Learning for Analysis of Complex Production Data in a Pre-Salt Carbonate Field Project.

2021-II **Invited Professor**, *National University of San Agustin*, Design and Analysis of Algorithms.

2021-I **Invited Professor**, *National University of San Agustin*, Artificial Intelligence.

2021-I **Virtual Professor**, *Data Science Research Peru*, Introduction to Deep Learning.

2020-II **Invited Professor**, *National University of San Agustin*, Design and Analysis of Algorithms.

2020-II **Virtual Professor**, *Data Science Research Peru*, Introduction to Machine Learning.

2020-I **Collaborator Professor**, *State University of Campinas (UNICAMP)*, Campinas-Brazil.
Topics in Computer Graphics.

2018-I – 2019-II **Postdoctoral research fellow**, *State University of Campinas (UNICAMP)*, Campinas-Brazil. Supervisor: Dr. Anderson Rocha.
Data Analysis, Image Processing and Machine Learning in Similarity Exploration from 4D seismic data.

2017-II – 2018-I **Software Development Engineer - Back-end**, *Monitora Soluções Tecnológicas / VistaJet*, São Carlos-Brazil.
Development of VISTAJET Platform (global business aviation company). API Development, Java Development, Oracle SQL, Hibernate, Git, Spring MVC, SCRUM Methodology, React, Redux.

2015-II – 2016-I **Researcher Scholar-Junior Specialist Position**, *University of California, Davis - USA*, Institute for Data Analysis and Visualization (IDAV). Supervisor: Dr. Bernd Haman, **FAPESP fellowship**.
Visual Analytics, Information Visualization, Data Mining.

2014-II **Assistant Professor**, *ICMC-USP*, Course: Computational Visualization. Supervisor: Dr. Maria Cristina Ferreida de Oliveira.
Data analysis. Visualization techniques. Visualization taxonomies. Organization and data types in visualization. Volume visualization techniques. Surface-based volume visualization techniques. Vector visualization. Multidimensional data visualization. Visual data mining. Alternative techniques for data representation.

2014-I **Assistant Professor**, *ICMC-USP*, Course: Algorithms and Data Structures II. Supervisor: Dr. Maria Cristina Ferreida de Oliveira.
Fundamentals of files and secondary storage. Organizing files. Indexing and maintenance of indexed files. Co-sequential processing and external sorting. B-Trees and its variations. Data structures for graphs representation. Classic algorithms on graphs and applications.

2013-I **Assistant Professor**, *ICMC-USP*, Course: Introduction to Computer Science (Laboratory). Supervisor: Jo Ueyama.
Problem solving and algorithm design. Programming structures. Simple data types. Modularization. Composite data types. Files. Debugging. Structured programming language.

2010 **Fellowship in Information Technology**, *Southern Peru Copper Corporation*, Cuajone, Moquegua-Peru.
Support of mine systems and standard user software, installation and configuration of PCs and servers, network management, cabling, basic hardware maintenance, analysis and design, batch development, .NET development, database Management: Sql server 2005.

2009-I(2) **Junior Analyst Developer**, *Global System And Consulting*, Arequipa-Peru.
Development of System of Non Structured Queries - Project Financed by IDB. Platform j2ee, Struts, Spring, Quartz; Windows

2009-I(1) **Junior Analyst Developer**, *Global System And Consulting*, Arequipa-Peru.
Development of Web System for Module of Citizen Attention and Governmental Management - System for Mexican Government. Platform j2ee, Struts, Spring, Quartz; Windows

2007-II/08-I **Junior Analyst Developer - Internship in company**, *CDI Solutions*, Arequipa-Peru.
Development of Web System of Library management. Programming in PHP, JavaScript, HTML; Database management: PostgreSQL; Apache server, networks, and Internet services on Linux Platform.

## Workshop and Conference Reviewing

2022 **Track on Social Network and Media Analysis and Mining (SNMAM)**.

2022 **International Conference on Information Management and Big Data (SIMBig)**.

2021 **Simposio de Computación Gráfica, Realidad Virtual y Procesamiento de Imágenes (SLCGRVPI)**, *CLEI*.

2020 **Journal of Universal Computer Science**.

2020 **Track on Social Network and Media Analysis and Mining (SNMAM)**.

2020 **International Conference on Information Management and Big Data (SIMBig)**.

2020 **Simposio de Computación Gráfica, Realidad Virtual y Procesamiento de Imágenes (SLCGRVPI)**, *CLEI*.

2020 **Simposio Latinoamericano de Teoría Computacional**, *CLEI*.

2020 **Journal of Applied Acoustics**.

2019 **International Conference on Information Management and Big Data (SIMBig)**.

2019 **Journal of Universal Computer Science**.

2018 **Journal of Universal Computer Science**.

2018 **Simpósio Brasileiro em Segurança da Informação e de Sistemas Computacionais (SBSeg)**.

2018 **Simposio de Computación Gráfica, Realidad Virtual y Procesamiento de Imágenes (SLCGRVPI)**, *CLEI*.

2016 **Women in Machine Learning (WiML)**, *CLEI*.

2016 **Simposio de Computación Gráfica, Realidad Virtual y Procesamiento de Imágenes (SLCGRVPI)**, *CLEI*.

2015 **Simposio de Computación Gráfica, Realidad Virtual y Procesamiento (SLCGRVPI)**, *CLEI*.

2015 **WSCG 2015 – 23rd Conference on Computer Graphics, Visualization and Computer Vision**.

2014 **Encontro Nacional de Inteligência Artificial e Computacional (ENIAC)**.

## Session Chair

2018 **Simposio de Computación Gráfica, Realidad Virtual y Procesamiento de Imágenes (SLCGRVPI)**, *CLEI*.

## Evaluation Committee

2022 **Neural Architecture Search Analysis for Object Detection (Master Defense)**, *Student: Cicera Vanessa Marques Sampaio Sidrim*, Committee members: Dra. Sandra Eliza Fontes de Avila (IC/Unicamp), Dr. Levy Boccato (FEEC/Unicamp), Dra. Aurea Soriano Vargas (IC/Unicamp), Universidade de Campinas.

2021 **Importance Sampling Sub-path Connections on Bidirectional Light Transport (Master Qualifying Exam)**, *Student: João Vítor Buscatto Silva*, Committee members: Dr. Hélio Pedrini (IC/Unicamp), Dr. André Santanché (IC/Unicamp), Dra. Aurea Soriano Vargas (IC/Unicamp), Universidade de Campinas.

2021 **Visualização de vetores de texto por meio de projeções multi-dimensionais (Undergraduate Project Defense)**, *Student: Luís Felipe Braga Gebrim Silva*, Committee members: Dr. Vinícius Ruela Pereira Borges (CIC/UnB), Dra. Aurea Soriano Vargas (IC/Unicamp), Dr. Geraldo Pereira Rocha Filho (CIC/UnB), Universidade de Brasília.

2021 **Interpretabilidade em Linguística Computacional: Integração de Técnicas Aplicadas ao BERT (Master Qualifying Exam)**, *Student: Rafael Bezerra de Menezes Rodrigues*, Committee members: Dr. Danilo Medeiros Eler (UNESP), Dra. Aurea Soriano Vargas (UNICAMP), Dr. Daniel Carlos Guimarães Pedronette (UNESP), Universidade Estadual Paulista (UNESP) Câmpus de Presidente Prudente.

2021 **Importance Sampling Sub-path Connections on Bidirectional Light Transport (Master Qualifying Exam)**, *Student: João Vítor Buscatto Silva*, Committee members: Dr. Hélio Pedrini (IC/Unicamp), Dr. Afonso Paiva Neto (ICMC/USP), Dra. Aurea Soriano Vargas (IC/Unicamp), Universidade de Campinas.

2018 **XIII Workshop de Teses, Dissertações e Trabalhos de Iniciação Científica (WTD)**, State University of Campinas.

2018 **Reviewer at 2nd Simpósio Internacional de Iniciação Científica e Tecnológica da USP - SIICUSP**, University of São Paulo.

2017 **Estudo e Avaliação de Softwares de Fotogrametria**, *Student: Rita Raad*, Committee members: Maria da Graça Campos Pimentel and Aurea Rossy Soriano Vargas, ICMC- University of São Paulo.

## Talks

2022 Exploratory Data Analysis

*Santander Bank - Brazil*

*Data Science Research Perú*

2021 Artificial Intelligence and Machine Learning in Industry

*JAKU Business Incubator - National University of San Agustin (UNSA)*

2021 Anomaly Detection

*12a Semana da Engenharia, do Centro Universitário da Fundação Hermínio Ometto - FHO-Universidade de Araras*

2021 Forensics Analysis of Digital Data.

*ML/DL Meetup AQP*
*16 Congresso Científico "Desafios para Construção de um Mundo Sustentável", do Centro Universitário da Fundação Hermínio Ometto - FHO-Universidade de Araras*

2020 Anomaly Detection in Time Series Data.

*VI Semana de Tecnologia da Informação (SETI) - UFLA*
*Data Science Research Peru - Arequipa*

2020 Visualization for Machine Learning.

*EEP/FUMEP - Live dia 19/10 MACHINE LEARNING*
*I Encuentro de Mujeres Peruanas en Computación - University of La Salle*
*XI Peruvian Symposium on Artificial Intelligence- Catholic University of San Pablo*
*ML/DL Meetup AQP - Technological University of Peru*

2019 Exploring time-varying multivariate data through a visual analytics framework.

*Khipu 2019 -Latin American meeting in Artificial Intelligence*

2019 Machine Learning Techniques to Estimate Reservoir Property Changes directly from 4D Seismic Attributes and Rock Property Maps. *I Seminar on Artificial Intelligence Applied to the Petroleum Industry*

2019 Presentation attack detection.

*16th International SummerSchool on Biometrics*

2017 Visual analytics of time-varying multivariate ionospheric scintillation data.

*Conference on Graphics, Patterns and Images - (SIBGRAPI)*

2014 Visual classification of ionospheric scintillation data.)

*Workshop Calibra (MundoGeo # connect - LatinAmerica*

2014 Visualization of music collections based on structural similarity.

*Conference on Graphics, Patterns and Images - (SIBGRAPI)*

## Languages

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| English | Advanced | *Professional proficiency* |
| (Test of English for International Communication (TOEIC) - 775) - 2015 | | |
| Portuguese | Advanced | *Professional proficiency* |
| Spanish | Advanced | *Native* |
| Italian | Basic | *Novice* |

## Volunteer Experience

- (2019-2021) Mentor in Sisay Mentores, Perú.
- (2020) Mentor in Data Science Research, Perú.
- (2008) Volunteer in a Christian Life Movement, Arequipa, Perú.
- (2000-2002) Volunteer in a Salesian Sisters project, Mollendo, Perú.

## Publications

### Journals and Book Chapters

[1] Rafael Werneck, **Soriano-Vargas, Aurea**, Renato Moura, Pedro Mendes Júnior, Raphael Prates, Manuel Castro, Maiara Gonçalves, Manzur Hossain, Marcelo Zampieri, Alexandre Ferreira, Alessandra Davolio, Bernd Hamann, Denis Schiozer, and Anderson Rocha. Data-driven deep-learning forecasting for oil production and pressure. *Journal of Petroleum Science and Engineering*, pages 1–21, 2021.

[2] **Soriano-Vargas, Aurea**, Rafael Werneck, Renato Moura, Pedro Mendes Júnior, Raphael Prates, Manuel Castro, Maiara Gonçalves, Manzur Hossain, Marcelo Zampieri, Alexandre Ferreira, Alessandra Davolio, Bernd Hamann, Denis Schiozer, and Anderson Rocha. A visual analytics approach to anomaly detection in hydrocarbon reservoir time series data. *Journal of Petroleum Science and Engineering*, page 108988, 2021.

[3] Caroline Mazini, **Soriano-Vargas, Aurea**, Anderson Rocha, and Zanoni Dias. Manifold learning for real-world event understanding. *IEEE Transactions on Information Forensics and Security*, 2021.

[4] Klaus Rollmann, **Soriano-Vargas, Aurea**, Forlan Almeida, Alessandra Davolio, Denis José Schiozer, and Anderson Rocha. Convolutional neural network formulation to compare 4-d seismic and reservoir simulation models. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2021.

[5] Klaus Rollmann, **Soriano-Vargas, Aurea**, Marcos Cirne, Alessandra Davolio, Denis José Schiozer, and Anderson Rocha. A three-way convolutional network to compare 4d seismic data and reservoir simulation models in different domains. *Journal of Petroleum Science and Engineering*, page 109260, 2021.

[6] **Soriano-Vargas, Aurea**, Klaus Rollmann, Forlan Almeida, Alessandra Davolio, Bernd Hamann, Denis J Schiozer, and Anderson Rocha. A synthetic case study of measuring the misfit between 4d seismic data and numerical reservoir simulation models through the momenta tree. *Computers & Geosciences*, 145:104617, 2020.

[7] Luís A. M. Pereira, Allan Pinto, Fernanda A. Andaló, Alexandre M. Ferreira, Bahram L. Sefidgari, **Soriano-Vargas, Aurea**, Marcos V. M. Cirne, and Anderson Rocha. The rise of data-driven models in presentation attack detection. In Richard Jiang, editor, *Deep Biometrics*. Springer, 2019.

[8] **Soriano-Vargas, Aurea**, Maria Cristina F Oliveira, and Bernd Hamann. TV-MV analytics framework to explore time-varying multivariate data from three domains: Crime statistics, air quality and stock market. *Information Visualization Journal*, 2019.

[9] **Soriano-Vargas, Aurea**, Bruno C Vani, Milton H Shimabukuro, João FG Monico, Maria Cristina F Oliveira, and Bernd Hamann. Visual analytics of time-varying multivariate ionospheric scintillation data. *Computers & Graphics*, 68:96–107, 2017.

### Conferences and Workshops

[10] **Soriano-Vargas, Aurea**, Klaus Rollmann, Forlan Almeida, Alessandra Davolio, Denis J. Schiozer, Bernd Hamann, and Anderson Rocha. Leveraging phylogenetic trees to assess variability of reservoir models. *SPE Latin America and Caribbean Petroleum Engineering Conference*, 2020.

[11] Marcos Cirne, **Soriano-Vargas, Aurea**, Masoud Maleki, Shahram Danaei, Alessandra Davolio, Denis J. Schiozer, and Anderson Rocha. Machine learning techniques to estimate reservoir property changes directly from a 4d seismic attribute and geographic coordinates. *2nd SBGf & SEG ML Workshop on Machine Learning*, 2020.

[12] Oscar Cuadros Linares, **Soriano-Vargas, Aurea**, Bruno Faical, Bernd Hamann, Alexandre Fabro, and Agma J. M. Traina. Efficient segmentation of cell nuclei in histopathological images. *IEEE 33rd International Symposium on Computer Based Medical Systems (CBMS)*, 2020.

[13] **Soriano-Vargas, Aurea**, Masoud Maleki, Shahram Danaei, Alessandra Davolio, Denis J. Schiozer, and Anderson Rocha. Machine learning techniques to estimate reservoir property changes directly from 4d seismic attributes and rock property maps. *I Seminar on Artificial Intelligence Applied to the Petroleum Industry*, 2019.

[14] **Soriano-Vargas, Aurea**, Maria Cristina F Oliveira, and Bernd Hamann. Exploring time-varying multivariate data through a visual analytics framework - poster. *Khipu 2019 - Latin American meeting in Artificial Intelligence*, 2019.

[15] **Soriano-Vargas, Aurea**. Presentation attack detection - poster. *16th International Summer School on Biometrics*, 2019.

[16] **Soriano, Aurea** and Maria Cristina Ferreira de Oliveira. Visual classification of ionospheric scintillation data. *Workshop Calibra (MundoGeo # connect - Latin America)*, 2014.

[17] **Soriano, Aurea**, Fernando Paulovich, Luis Gustavo Nonato, and Maria Cristina Ferreira de Oliveira. Visualization of music collections based on structural similarity. *Conference on Graphics, Patterns and Images - (SIBGRAPI)*, 2014.

[18] Jorge Valverde-Rebaza, **Soriano, Aurea**, Lilian Berton, Maria Cristina Ferreira de Oliveira, and Alneu de Andrade Lopes. Music genre classification using traditional and relational approaches. *Brazilian conference on Intelligent systems (BRACIS)*, pages 259–264, 2014.

***The last update was performed on August 24th, 2022.***